Q-scan morphology of scattered-light transients in LIGO O3 and O4a

with an auxiliary-channel correlation analysis

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Abstract. Scattered light is among the most persistent sources of transient noise in the Advanced LIGO detectors. Stray light reflecting from a moving surface and rejoining the main beam is phase modulated by $4\pi x/\lambda$, and when that excursion exceeds a radian the noise appears in the time–frequency plane as a train of harmonic *arches* whose instantaneous frequency follows $f_n = 2n|v|/\lambda$. Because the frequency depends on the magnitude of the velocity, each cycle of the driving motion produces two arches, so the arch repetition period is a direct measurement of the mechanical disturbance responsible.

This work characterises the morphology of scattered-light transients in the third and fourth LIGO observing runs using only public data. A labelled O3 sample is drawn from the Gravity Spy classification release; every trigger is Q-transformed and reduced to a twenty-element feature vector describing the arch train. The O3 sample is then tied to physical cause in two ways: by rank-correlating the hourly trigger rate against seismometer band-limited RMS in the earthquake, microseismic and anthropogenic bands, and by fitting the $f_n = 2n|v|/\lambda$ predictor, trigger by trigger, to the Livingston OM3 suspension witness that the public O3 auxiliary release makes available at 256 Hz.

Two properties of the public data determine the shape of the O4a half of the analysis and are reported as findings in their own right: no Gravity Spy classification catalogue has been released for O4a, and the O4a auxiliary release contains no seismometer, accelerometer or suspension witness of any kind. The O4a sample is therefore constructed from strain alone, using an excess-power trigger finder and a classifier trained on the O3 features, and the O3 comparison is made like-for-like by running the identical search on both runs. The auxiliary analysis, being an O3 result, is what licenses reading the O4a arch-period distribution as a statement about ground motion.

Keywords: gravitational waves; LIGO; detector characterisation; transient noise; scattered light; time–frequency analysis.

Analysis code and configuration: https://github.com/hardiksingh/LIGO_Glitch_Classification

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1 Introduction

1.1 Transient noise and why its shape matters

The Advanced LIGO detectors [1] reach strain sensitivities of order $10^{-23}\,\mathrm{Hz}^{-1/2}$, and at that level the data are not Gaussian. Short bursts of excess power — *glitches* — occur at rates of order one per minute, and they matter for two reasons: they reduce the livetime available for astrophysical search, and they can mimic or corrupt real signals [2]. Detector characterisation exists to find them, explain them, and where possible remove their cause from the instrument.

The productive question about a glitch is rarely "how loud was it" but "what shape was it". Glitch classes are defined morphologically, in the time–frequency plane, and the morphology is what points back to a mechanism. The Gravity Spy project [3, 4] formalised this by combining citizen-science labelling with a convolutional classifier over 22 morphological classes, and its classification release for the first three observing runs is now the standard labelled dataset for this kind of work.

1.2 Scattered light

Among those classes, *scattered light* is unusual in that its physical mechanism is understood in closed form.

A small fraction of the circulating beam leaves the main path, strikes a nearby surface — a baffle, a suspended optic, a viewport — and rejoins the beam. Having traversed the extra path twice, the recombining field has acquired a phase

$$\phi(t) = \frac{4\pi\, x(t)}{\lambda}, \tag{1}$$

where x is the displacement of the scattering surface along the beam and $\lambda = 1064\,\mathrm{nm}$ is the Advanced LIGO laser wavelength. When ϕ stays well below a radian the coupling is linear and the scattered light simply adds a small amount of noise at the frequency of the motion. When ϕ exceeds a radian — which requires only a fraction of a micron of displacement — the modulation is strongly non-linear, and the noise appears at the instantaneous frequency

$$f_1(t) = \frac{1}{2\pi}\left|\frac{\mathrm{d}\phi}{\mathrm{d}t}\right| = \frac{2}{\lambda}\,|v(t)|, \tag{2}$$

with $v = \mathrm{d}x/\mathrm{d}t$. Numerically, $2/\lambda = 1.88 \times 10^6\,\mathrm{Hz/(m/s)}$: every micron per second of surface velocity produces 1.88 Hz of noise frequency.

Light that makes n round trips between the two surfaces before rejoining accumulates n times the phase, so the noise appears simultaneously at every harmonic

$$f_n(t) = n\,\frac{2}{\lambda}\,|v(t)|, \qquad n = 1, 2, 3, \ldots \tag{3}$$

with amplitudes falling off as the bounce order rises. In a Q-scan this produces the characteristic picture: a comb of smooth, harmonically spaced ridges, all peaking at the same instant.

Two consequences of Eq. (3) carry the whole analysis.

The arch period measures the driving motion. Because f_n depends on $|v|$ rather than on v, it reaches a maximum twice per cycle of the motion. The observed arch repetition period is therefore *half* the period of the disturbance,

$$f_{\mathrm{drive}} = \frac{1}{2\,T_{\mathrm{arch}}}. \tag{4}$$

A measured T_{arch} of 3.3 s implies a driver at 0.15 Hz, squarely in the microseismic band; a measured 0.25 s implies 2 Hz, in the anthropogenic band. The morphology alone, with no auxiliary data at all, identifies the class of disturbance.

The harmonic spacing measures the fundamental. At the apex of an arch train, all harmonics peak together, so a frequency slice through the apex is a comb with spacing $f_1^{\max} = 2v_{\mathrm{peak}}/\lambda$. The spacing is thus a direct readout of the peak surface velocity, again from strain alone.

1.3 What is already known

Soni et al. [5] characterised scattering throughout O3 and identified two distinct populations at LIGO: a *slow* component driven by the 0.1 to 0.3 Hz microseism, whose arches recur every few seconds and reach a few tens of hertz, and a *fast* component driven by 1 to 6 Hz motion, whose arches recur several times per second. They traced the dominant O3 slow-scattering coupling to relative motion of a few microns between the main suspension chain and its reaction chain, and demonstrated mitigation by damping that relative motion. A witness-channel algorithm predicting when scattering should be present is a central result of that work, and Eq. (3) is its basis.

Related work has established the surrounding picture: environmental coupling paths in general [6], scattering from end benches at Virgo [7, 8], and the O3 data-quality products [2]. For the fourth run, Capote et al. [9] document the instrument performance and Soni et al. [10] the detector characterisation of its first half. Most directly relevant, Nandi et al. [11] report on scattered light at Livingston during O4 and again find two populations, one traced to microseismic ground motion and mitigated by baffle installation, the other to vertical ground motion coupling through a vacuum chamber.

That body of work is, however, built on internal LIGO data: full auxiliary channel sets, engineering records, and access to the instrument. The present study asks a narrower and quite different question.

1.4 This work

How much of the scattered-light story can be reconstructed from public data alone, and does the answer change between O3 and O4a?

The question is worth asking for two reasons. It is a genuine test of what the open data releases support, which is useful to anyone outside the collaboration. And it forces every inference to be

made from quantities that can be independently checked, which is a discipline that internal-data analyses do not always impose.

The analysis proceeds in five steps.

1. A labelled O3 sample of scattered-light triggers is drawn from the Gravity Spy release [12], together with a control sample of other glitch classes.
2. Each trigger is Q-transformed [13] and reduced to a twenty-element morphology feature vector, described in Section 3, whose components are the physical quantities of Eqs. (3) to (4) rather than generic image statistics.
3. The O3 sample is tied to physical cause using the public auxiliary channels: the hourly trigger rate is rank-correlated against seismometer band-limited RMS, and the predictor of Eq. (3) is fitted trigger by trigger to a suspension witness.
4. A classifier trained on the O3 features is applied to a livetime-limited blind search of O4a strain.
5. The two populations are compared in rate, in slow/fast composition, and feature by feature.

Steps 3 and 4 are shaped by two properties of the public releases that were established before any code was written and that are reported here as results in their own right, because they are not stated plainly anywhere else:

- **No Gravity Spy classification catalogue exists for O4a.** The Zenodo release [12] covers O1 through O3b. The O4a Gravity Spy model is described by Wu et al. [14] but no catalogue accompanies it. Independent confirmation comes from Cirfeta et al. [15], who work on O4a without labels and use the O3b Gravity Spy set as their reference index — exactly the workaround that O4a labels would render unnecessary.
- **The O4a auxiliary release contains no motion witness.** Its 25 channels [16] are calibration lines, input mode-cleaner wavefront sensors, photon-calibrator readouts, mains monitors and parametric-instability monitors. There is no seismometer, accelerometer, microphone or suspension channel, so neither the rate correlation nor the per-trigger predictor can be computed for O4a from public data.

The O3 auxiliary release, by contrast, proved richer than expected: 86 full-rate channels and 402 second-trend channels [17], including the complete ground-seismometer band-limited RMS family across the earthquake, microseismic and anthropogenic bands at both sites, and — crucially — `L1:SUS-OM3_M1_DAMP_L_IN1_DQ` at 256 Hz, a genuine optic-motion witness and the only full-rate one in either release.

Section 2 sets out the data and verifies its availability; Section 3 the analysis; Section 4 the validation against simulated injections with known parameters; Section 5 the results; Section 6 their interpretation and limits.

2 Data

Every dataset used here is public. Nothing in the analysis requires LIGO credentials. All availability statements below were verified against the live services on 2026-08-04 by the environment-check stage of the pipeline (Appendix B).

2.1 Observing-run epochs

Table 1 lists the epochs analysed and their GPS boundaries, taken from the GWOSC data-release pages.

Table 1: Observing-run epochs. Strain from both detectors is public for all three. O4b and O4c are excluded: their releases were partial at the time of writing and they add nothing to the O3-versus-O4a comparison.

Epoch	Run	GPS start	GPS end	UTC span
O3a	O3	1238166018	1253977218	2019-04-01 to 2019-10-01
O3b	O3	1256655618	1269363618	2019-11-01 to 2020-03-27
O4a	O4	1368975618	1389456018	2023-05-24 to 2024-01-16

Strain is taken from the 4 kHz release, which is sufficient for an analysis confined below 256 Hz. Analysis-ready segments come from the GWOSC segment API using the `<IFO>_DATA` flag; this is the correct denominator for a glitch rate, since a rate computed against calendar time would confound glitchiness with duty cycle.

2.2 Gravity Spy classifications: O3 only

The labelled input is the Gravity Spy machine-learning classification release, Zenodo record 5649212 [4, 12]. It contains every Omicron [18] trigger with SNR > 7.5 and peak frequency in 10 to 2048 Hz, together with the trigger metadata, a 22-element vector of class posteriors, and the arg-max label.

Selection cuts, applied identically to the target and control classes so that both are drawn under the same conditions:

- SNR ≥ 7.5 (the Gravity Spy trigger threshold);
- peak frequency in 10 to 2048 Hz;
- machine-learning confidence in the assigned class ≥ 0.90;
- event time inside the epoch boundaries of Table 1.

The confidence cut is the one real choice here. It trades sample size for label purity, and at 0.90 it removes the ambiguous tail without collapsing the sample. Section 5.6 reports how the results move when it is varied.

The control sample consists of the classes *Blip*, *Koi_Fish*, *Low_Frequency_Burst*, *Tomte*, *Extremely_Loud*, *Repeating_Blips*, *Light_Modulation*, *Violin_Mode*, *Whistle*, *Power_Line* and *No_Glitch*. Its role is to define what the classifier of Section 3.7 must reject.

There is no O4a catalogue. This is worth stating precisely because the analysis design turns on it. The Zenodo ML-classification release covers O1, O2, O3a and O3b. The volunteer-classification release extending to 2024-07-20 is keyed by Zooniverse subject identifier and carries no GPS time, detector or trigger metadata, so it cannot serve as a trigger list. The O4a Gravity Spy model of Wu et al. [14] is published without an accompanying catalogue. Consequently the O4a sample must be built from strain (Section 3.8).

2.3 Auxiliary channels

Both GWOSC auxiliary releases are served over NDS2 at `nds.gwosc.org`. Only short windows around trigger times, and hour-scale trends, are needed here, so the several-terabyte bulk downloads are unnecessary.

2.3.1 O3

The O3 release [17] comprises 86 full-rate channels (40 H1, 46 L1) and 402 second-trend channels at 1 Hz (190 H1, 212 L1). Table 2 lists those the analysis uses.

Table 2: Auxiliary channels used in the O3 analysis. Names ending `.rms` come from the second-trend frames. The suffix `BLRMS_100M_300M` denotes a band-limited RMS over 0.1 to 0.3 Hz; "M" is millihertz.

Channel	Rate	Role
`L1:SUS-OM3_M1_DAMP_L_IN1_DQ`	256 Hz	OM3 top-stage *length* damping error signal; the motion witness for the per-trigger predictor
`[HL]1:ISI-GND_STS_*_BLRMS_100M_300M.rms`	1 Hz	microseismic band at ETMX, ETMY, ITMX, ITMY, HAM5 in x, y, z; driver of slow scattering
`[HL]1:ISI-GND_STS_*_BLRMS_{1_3,3_10}.rms`	1 Hz	1 to 3 Hz and 3 to 10 Hz anthropogenic bands; driver of fast scattering
`[HL]1:ISI-GND_STS_ITMY_Z_BLRMS_30M_100M`	16 Hz	0.03 to 0.10 Hz earthquake band

Two asymmetries are worth recording. First, L1 has a full-rate suspension witness in the public release and H1 does not; the per-trigger predictor of Section 3.6 is therefore an L1-only result, and the H1 auxiliary analysis is confined to the rate level. Second, the auxiliary channels are published in raw digital counts with no calibration to metres, which is dealt with in Section 3.6.

2.3.2 O4a

The O4a release [16] is 25 channels: 14 from H1 and 11 from L1. They are reproduced in full below, because the point being made depends on the list being complete rather than summarised.

```
H1: CAL-PCALX_RX_PD_OUT_DQ        L1: ASC-X_TR_A_NSUM_OUT_DQ
    CAL-PCALY_RX_PD_OUT_DQ            CAL-PCALX_RX_PD_OUT_DQ
    IMC-WFS_A_DC_PIT_OUT_DQ           CAL-PCALY_RX_PD_OUT_DQ
    IMC-WFS_A_DC_YAW_OUT_DQ           IMC-WFS_B_I_PIT_OUT_DQ
    LSC-POP_A_LF_OUT_DQ               OAF-IMC_WFS_B_I_PIT_PREFILT_OUT_DQ
    LSC-REFL_A_RIN_OUT_DQ             PEM-EX_VMON_ETMX_ESDPOWER24_DQ
    OAF-IMC_WFS_A_DC_PIT_PREFILT_OUT_DQ PEM-EY_MAINSMON_EBAY_1_DQ
    OAF-IMC_WFS_A_DC_YAW_PREFILT_OUT_DQ SUS-ETMX_L1_CAL_LINE_OUT_DQ
    OAF-REFL_A_RIN_PREFILT_OUT_DQ     SUS-ETMX_L2_CAL_LINE_OUT_DQ
    PEM-EY_MAINSMON_EBAY_1_DQ         SUS-ETMX_L3_CAL_LINE_OUT_DQ
    SUS-ETMX_L1_CAL_LINE_OUT_DQ       SUS-PI_PROC_COMPUTE_MODE5_RMSMON
```

```
SUS-ETMX_L2_CAL_LINE_OUT_DQ
SUS-ETMX_L3_CAL_LINE_OUT_DQ
SUS-PI_PROC_COMPUTE_MODE29_RMSMON
```

Every one is a photon-calibrator readout, an input mode-cleaner wavefront sensor, a length- or angular-sensing readout, a mains voltage monitor, an electrostatic-drive voltage monitor, a calibration-line drive, or a parametric-instability RMS monitor.

There is no seismometer, no accelerometer, no microphone and no suspension witness. Neither the rate-level correlation against ground motion nor the per-trigger scattering predictor is computable for O4a from public data. This is a limitation of the release, not of the method, and it is the reason the O3-versus-O4a comparison in Section 5.5 is a morphology comparison rather than a mechanism comparison.

The analysis pins this fact in code: a unit test asserts that the O4a channel list contains none of the tokens `ISI-GND`, `_SEIS_`, `_ACC_`, `SUS-OM` or `WIT_L`. Should GWOSC extend the release, that test fails, which is the intended signal to revisit the design.

2.4 Summary of what is and is not possible

Table 3: Analysis coverage by run, given the public data.

Analysis	O3	O4a
Q-scan morphology	yes, on Gravity Spy labels	yes, on classifier output
Rate versus ground motion	yes	**no** — no seismic channels
Per-trigger predictor	L1 only	**no** — no motion witness
Slow/fast decomposition	yes	yes, from arch period alone

The slow/fast decomposition survives into O4a precisely because Eq. (4) makes it a strain-only measurement. That is what makes the comparison possible at all, and it is the reason the auxiliary work in Section 5.3 matters beyond O3: it is the calibration that justifies reading an O4a arch period as a statement about ground motion.

3 Method

3.1 Q-transform maps

Each trigger is analysed on a normalised-energy Q-transform map [13]. The transform tiles the time–frequency plane with constant-Q bisquare windows and reports, for each tile,

$$Z = \frac{|X|^2}{\langle |X|^2 \rangle}, \tag{5}$$

the ratio of tile energy to the mean over tiles at the same frequency. For stationary Gaussian noise Z is exponentially distributed with unit mean, so $\sqrt{2Z}$ is the single-tile signal-to-noise ratio. The implementation is `gwpy`'s `q_transform` [19], which is the same algorithm used by the LIGO detector-characterisation tools; the standard `gwdetchar` wrapper is not used because its dependency chain has no Windows build, and nothing beyond the transform itself is needed.

64 s of strain is fetched around each trigger and only the central 16 s is analysed. The padding is not decorative: the whitening filter and the constant-Q filters both ring, and the resulting transients are exactly the kind of structure a ridge tracker would mistake for an arch. The configuration enforces that the padding exceeds the analysed window.

Two parameters were set by validation rather than by convention, and both mattered.

The Q range is capped at 16. `gwpy` selects the Q plane of maximum energy. Left unconstrained it selects $Q \approx 45$ for a scattering train, because a high-Q plane integrates the quasi-periodic structure coherently and reports enormous energy. But a tile of quality factor Q at frequency f lasts $Q/2\pi f$ seconds — 0.16 s at $Q = 45$ and 45 Hz — which is comparable to the repetition period of fast scattering. The arches are then unresolved. On a simulated fast event with an injected arch period of 0.250 s, the uncapped transform returned 4.99 s: wrong by a factor of twenty. With the cap it returns 0.250 s.

The ridge threshold is $Z = 20$, not $Z = 5.5$. A threshold has to sit above the largest energy a *noise-only* map produces, not merely above its mean. A map of this size holds of order 10^4 independent tiles, whose expected maximum is $\ln(10^4) \approx 9$ before the multi-Q search is taken into account and ≈ 16 after. At $Z = 5.5$ — a natural-looking choice, corresponding to single-tile SNR 3.3 — a pure-noise map produced hundreds of supra-threshold bins and the measured maximum arch frequency saturated at the top of the analysis band. $Z = 20$ (single-tile SNR 6.3) clears the noise; a connected-component filter removes the residue.

3.2 Ridge extraction

From the thresholded map, three tracks are formed per time bin: the frequency of the loudest tile, the highest supra-threshold frequency, and the lowest. The second of these — the *arch envelope* — is the quantity the auxiliary predictor predicts, and is what the morphology features describe.

Connected supra-threshold regions smaller than 40 output pixels are discarded. Because the output grid is interpolated from far coarser Q tiles, a single hot tile already spans tens of pixels, so this is a speck filter rather than a significance cut; the significance is carried by the threshold.

3.3 Morphology features

Each map is reduced to the twenty features listed in Appendix A. They fall into five groups: extent (peak, maximum and minimum frequency, bandwidth, duration, SNR); arch structure (count, repetition period, period scatter); harmonic structure (count and spacing at the apex); shape (RMS slope and curvature of the envelope, ratio of rising to falling time); and energy distribution (fraction below 40 Hz, fraction above 100 Hz, spectral centroid, band concentration, time symmetry).

They are computed on the normalised-energy map, so they are insensitive to the absolute calibration and directly comparable between detectors and between runs. Where a feature is undefined — `arch_period` when fewer than two arches are found — it is returned as `NaN` rather than as a fabricated value, and the classifier treats the missingness as informative.

Two of them deserve comment.

Arch period, from the autocorrelation. The obvious estimator is the median spacing between detected envelope maxima. It fails in exactly the case that matters: when the topmost harmonic dips below threshold the envelope drops to the harmonic below, creating a spurious local maximum that splits one arch into two. On a simulated slow event with an injected period of 3.33 s, peak spacing returned 2.80 s — 16 % low. The autocorrelation of the envelope is insensitive to this, because the periodicity is present in every harmonic simultaneously; it returns 3.36 s. The peak count is still reported separately, as `n_arches`.

Maximum arch frequency is biased high, and stays that way. A Q tile has bandwidth f/Q, so the supra-threshold region extends above the true apex of the arch by roughly half of that. Measured against injections, the bias is about 15 %. It is not tuned away: it is measured, asserted in the test suite, and quoted with the result. Since it applies identically to both runs it cancels in the O3-versus-O4a comparison, which is where the feature is actually used.

3.4 From triggers to events, and from events to blocks

Two properties of the real data, neither visible until the catalogue and the strain server were in hand, determine how the sample is built. Both are recorded here because each one changes a number that would otherwise be reported.

The catalogue lists triggers, not events. Gravity Spy publishes Omicron triggers. After the cuts of Section 2.2, 12.3 % of rows share an exact (event time, detector) pair – the same trigger under several Gravity Spy subject identifiers, one per Q-scan duration view – and about half of all triggers fall within 8 s of another one, which is to say inside each other's analysed Q-scan window. Those triggers therefore produce *identical* feature vectors.

Left alone, every morphology distribution would be weighted by how many Omicron triggers a given arch train happened to generate: loud, long trains counted dozens of times, quiet ones once. That is a selection effect acting directly on the axis this study measures. Worse, it is class-dependent. Clustering at 16 s – the full width of the analysed window, so that no two retained events can share a Q-scan – yields 2.44 triggers per event for scattered light but only 1.35 for the control classes. Uncorrected, scattering would have been over-represented by some 80 % relative to the controls, biasing both the classifier's training set and every rate comparison. The same window is applied to the O4a search output, so the two runs count the same kind of thing.

The unclustered trigger list is retained as well: the rate correlation of Section 3.5 legitimately uses *trigger* rate, and each result states which of the two it quotes.

The unit of cost is the file, not the event. GWOSC serves open strain in HDF5 files of 4096 s, and a request for any sub-interval downloads the whole containing file: two to four minutes on a domestic connection, essentially independent of how much was asked for. A 64 s Q-scan window therefore costs one full download, and a time-stratified sample of 4800 individual events touched 3484 distinct files – about 1.4 events per download, or three days of wall clock.

The analysis is restructured around that. Windows are cut from whole blocks held in a small in-memory cache, and the sample is drawn as *blocks first, events within blocks second*: 100 blocks per detector per epoch, stratified across each epoch's span, with up to 12 events taken at random from each. That gives 2256 scattered-light events from 400 blocks, and the measured yield in a production run is about ten events per download. The control sample is drawn from the *same* blocks, which costs no extra downloads and removes a confound at the same time: signal and background then come from identical stretches of data, so the classifier cannot separate the two classes by learning what a noisy night looks like.

The price is real and is carried into the interpretation. Events drawn from one block share ground-motion conditions and are therefore correlated, so the effective sample size is smaller than the row count: it is the *block* count, not the event count, that carries the statistical weight. Events are taken at random within a block rather than by loudness, which would bias the sample towards the tail.

3.5 Rate-level auxiliary correlation

Triggers are binned into GPS hours and divided by the analysis-ready livetime in each hour. Hours with zero livetime and hours with zero triggers are different things: the former must be dropped, the latter kept as genuine zeros. Conflating them is a reliable way to manufacture a spurious correlation with anything that tracks duty cycle, ground motion included.

Each hourly rate series is rank-correlated (Spearman ρ) against the hourly median band-limited RMS of every configured seismometer channel. The median within an hour is used rather than the mean because the trend channels contain occasional large excursions — earthquakes, calibration activity — whose effect on an hourly mean is out of proportion to how much of the hour they occupy.

Fetching those trends has its own constraints, measured against the live server rather than assumed. The GWOSC NDS host serves a four-hour request but refuses eight; nine channels requested together cost 11.3 s against 40.8 s one at a time, so requests are batched; and requests are issued only for chunks that overlap analysis-ready segments, since a chunk with no livetime contributes nothing to a rate correlation. Availability looked poor when probed at fixed calendar offsets and turned out to be complete – 18 of 18 – when probed uniformly in livetime, the earlier failures having been non-observing time.

Confidence intervals come from a moving-block bootstrap [20] with a six-hour block. This matters: the microseism varies on a timescale of days, so hourly rates are strongly autocorrelated and the naive standard error on ρ is far too small. Three further checks guard against the standard failure modes:

- a **quantile table**, giving the mean rate in each quartile of the driving channel and the top-to-bottom ratio, so that the result is stated in units a commissioner can act on rather than as a correlation coefficient;
- a **lag scan** over ± 12 h. Scattering responds to ground motion essentially instantaneously,

so the correlation must peak at zero lag. A peak at large lag would indicate that both series are being driven by a slow common trend — diurnal or weather — rather than by the mechanism under test;

- a **partial correlation**, testing whether an apparent correlation with the 1 to 6 Hz anthropogenic band survives once the microseism is controlled for.

3.6 Trigger-level scattering predictor

For L1, where a full-rate motion witness is public, the predictor of Eq. (3) is tested trigger by trigger.

The witness is band-passed over 0.03 to 10 Hz — the lower corner removes the DC drift that differentiation would otherwise amplify into a ramp — and differentiated to give velocity. Harmonics $n = 1 \ldots 8$ of Eq. (3) are then compared with the measured arch envelope. A ridge point at frequency f counts as *explained* if some harmonic lies within $\max(1.5\,\mathrm{Hz}, 0.15 f)$ of it, and the headline statistic is the fraction of supra-threshold time bins that are explained.

The one free parameter. GWOSC publishes auxiliary channels in raw counts with no calibration to metres. Rather than invent one, a single scale factor α (metres per count) is fitted per detector and epoch by maximising the mean explained fraction over a training set of triggers, using a logarithmic grid over eight decades followed by a golden-section refinement. It is then *frozen* and applied to a disjoint validation set, on which the quoted agreement is measured. The width of the range over which the objective stays within 1 % of its maximum is reported alongside: a narrow range means the fit is well constrained, a wide one is a warning that the witness does not in fact track the scattering surface.

Section 4 shows this procedure recovering a known calibration from a synthetic witness to 7 %.

3.7 The strain-only identifier

To carry the O3 definition of the class into O4a, a gradient-boosted tree classifier [21, 22] is trained on the O3 features to separate *Scattered_Light* from the control classes. Missing features are median-imputed with a missingness indicator, so that "no arch period could be measured" is available to the model as information rather than being silently filled in.

Performance is measured by stratified five-fold cross-validation, and the operating point is fixed at a false-positive rate of 0.01 against the control sample, then frozen.

A third constraint is arithmetic and binds harder than either of the caveats below. An operating point can only be placed as finely as the background sample resolves: at a false-positive rate p with n control events per fold, the threshold is fixed by $p\,n$ expected false positives. Trained on a trial sample of 66 controls, the cross-validated AUC was a stable 0.919 ± 0.033 while the efficiency at $p = 0.01$ came out 0.368 ± 0.334, with individual folds returning 0.000 (at an infinite threshold), 0.160, 0.680, 0.760 and 0.240 — because $p\,n = 0.13$, so the threshold was decided by whether zero or one event happened to land above it. The code now computes $p\,n$, warns below five, and

records the fact in the saved model; the production control sample gives $p\,n \approx 7$ and clears it. The AUC is threshold-free and unaffected throughout.

Two further caveats travel with those numbers and are repeated wherever they are quoted. The efficiency and false-positive rate are measured against the *Gravity Spy control classes*, so they say how well the model separates scattered light from other known O3 glitch classes; they do not and cannot measure performance against whatever unlabelled O4a contains. And the folds are not grouped in time, so triggers minutes apart can land in different folds, making the quoted AUC mildly optimistic. The sample is drawn stratified across the whole run, which makes such neighbours rare but does not eliminate them.

3.8 The O4a search

With no O4a trigger catalogue, triggers are generated from strain. The statistic is the 0.25 s RMS of whitened, band-passed (10 to 120 Hz) strain, converted to a robust z-score using the median and median-absolute-deviation of the surrounding stretch. Robust statistics are essential here: during a scattering-heavy hour a large fraction of the data is glitching, and a mean/standard-deviation normalisation would be dragged upward by the very signal it is meant to detect. Candidates are clustered with a 4 s window, since an arch train is one physical event lasting seconds, and candidates within the whitening-filter transient at either end are discarded.

This is a deliberately loose stand-in for Omicron: its job is to lose no scattering, not to be pure. Purity is the classifier's job.

Livetime, not calendar time. A full O4a search is out of reach on the hardware available. The search draws 1200 windows of 512 s per detector, sampled uniformly in livetime — about 7.1 d each, roughly 4 % of the run. All rates are therefore quoted per hour of *analysed* livetime, and the coverage fraction is recorded with every result.

Like-for-like comparison. The identical search is also run on O3b. This is what makes the comparison interpretable: with the same trigger generator and the same classifier on both sides, a difference in rate or morphology cannot be attributed to the change from the Gravity Spy pipeline to this one. The comparison against the Gravity Spy O3 sample is also reported, since it has better labels; where the two agree the result is robust, and where they disagree the difference is a selection effect and is reported as such.

3.9 Population statistics

Rates are Poisson counts over a known livetime, so intervals are exact (Garwood) Poisson intervals rather than $\sqrt{N}$.

Feature distributions are compared with the two-sample Kolmogorov–Smirnov test for the p-value and with Cliff's δ [23] for the effect size. With tens of thousands of triggers per sample essentially every KS test rejects, so the p-value carries little information; the effect size is what carries the physics and is quoted first. Because many features are tested at once, p-values are corrected by the Benjamini–Hochberg procedure [24] and the corrected values are the ones reported.

The headline slow-fraction change is a difference of two binomial proportions, reported with a Newcombe score-based interval [25], which behaves correctly as a proportion approaches 0 or 1 where the Wald interval does not.

4 Validation against known signals

Every number in Section 5 is produced by an algorithm that could, in principle, be measuring an artefact of its own construction. The only way to tell is to give it a signal whose parameters are known exactly and check what comes back. This section does that, and it is the reason two of the analysis choices in Section 3 are what they are.

4.1 The injection model

The simulator generates

$$x(t) = X_0 \sin(2\pi f_m t + \varphi), \tag{6}$$

$$h(t) = \sum_{n=1}^{N} K\, r^{\,n-1} \sin\!\left(n\, \frac{4\pi\, x(t)}{\lambda} \right), \tag{7}$$

where the sum runs over bounce orders with amplitude ratio r between successive bounces. The arches are therefore not drawn on: they emerge from the phase modulation, as they do in the instrument. From Eqs. (6) to (7) the expected measurements follow:

$$f_1^{\max} = \frac{4\pi f_m X_0}{\lambda}, \qquad f^{\max} = N f_1^{\max}, \qquad T_{\mathrm{arch}} = \frac{1}{2 f_m}, \tag{8}$$

and the harmonic comb at the apex has spacing $f_1^{\max}$.

A wrong model, and what it cost. The first version of the simulator used a single term, $h = K \sin(4\pi x/\lambda)$, on the reasoning that the Jacobi–Anger expansion of a sine of a large argument produces harmonics automatically. It does — but they are harmonics of the *motion* frequency f_m, and they terminate at $k \approx \beta$ where $\beta = 4\pi X_0/\lambda$ is the modulation index. Since βf_m is exactly $f_1^{\max}$, a one-bounce model produces *one* arch however large β is. Raising the displacement to "get more harmonics" only makes that single arch taller. The harmonic arches of a real Q-scan come from multiple round trips, which is what Eq. (7) encodes.

A second false start is worth recording because it recurs in real analyses. A scattering train that fills the entire analysed window is whitened away completely: the transform estimates its amplitude spectral density from the data it is given, and a signal occupying the whole stretch *is* the noise floor as far as that estimate is concerned. The injection is therefore confined to a Tukey-windowed stretch occupying a modest fraction of the window, as a real transient is.

4.2 Recovery

Two canonical injections are used throughout: a *slow* event ($X_0 = 8\,\mu\mathrm{m}$, $f_m = 0.15\,\mathrm{Hz}$, four bounces) and a *fast* event ($X_0 = 0.64\,\mu\mathrm{m}$, $f_m = 2\,\mathrm{Hz}$, three bounces), both at unit-variance white

noise. Table 4 gives the recovery.

Table 4: Recovery of injected parameters. The witness calibration is fitted by the procedure of Section 3.6 without being told the answer; the synthetic witness was generated at 10^9 counts per metre, so the value to be recovered is $1.00 \times 10^{-9}\,\mathrm{m\,count^{-1}}$.

	Slow injection		Fast injection	
Quantity	injected	recovered	injected	recovered
Arch period T_{arch} (s)	3.33	3.36	0.250	0.250
Harmonic spacing (Hz)	14.2	13.9	15.1	14.0
Arch count	~ 5	6	~ 32	31
Maximum arch frequency (Hz)	56.7	65.8	45.4	62.8
Witness calibration (10^{-9} m/count)	1.00	1.07	1.00	1.09
Envelope explained by predictor	—	95 %	—	84 %
Predictor–envelope correlation r	—	0.92	—	0.83

The arch period, which carries the physical interpretation through Eq. (4), is recovered to 0.8 % for the slow injection and exactly for the fast one. The harmonic spacing, which measures the peak surface velocity, is recovered to within 7 %. The witness calibration — fitted blind, over an eight-decade search range — comes back within 9 %.

The maximum arch frequency reads systematically high, by 16 % for the slow injection and 38 % for the fast one, for the reason given in Section 3.3: a Q tile of bandwidth f/Q extends above the true apex. The bias is larger for the fast injection because its arches are shorter-lived relative to the tile duration. This feature is therefore not used for any absolute statement; it is used only in the O3-versus-O4a comparison, where the bias applies identically on both sides.

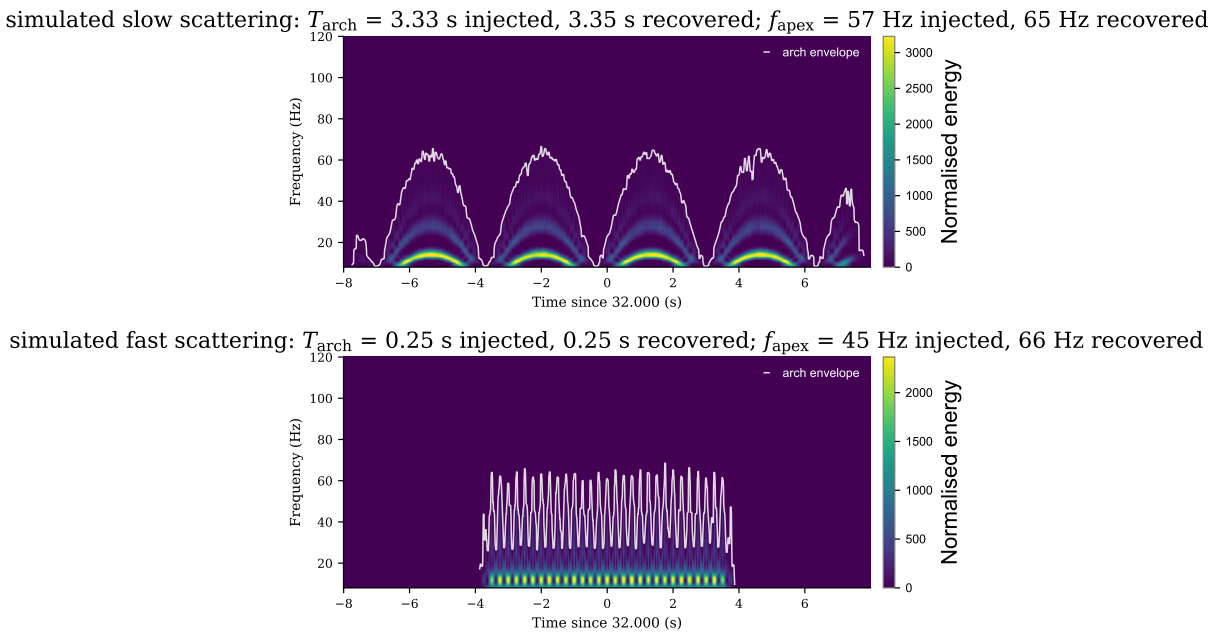


Figure 1: Q-scans of the two canonical injections with the extracted arch envelope overlaid (white). Upper: slow scattering, 0.15 Hz drive, arches recurring every 3.3 s. Lower: fast scattering, 2 Hz drive, arches recurring every 0.25 s. Colour is normalised energy Z; the ridge threshold is $Z = 20$. Produced by `scripts/07_make_figures.py -only validation`.

4.3 The null test

A threshold that lets noise through invalidates every extent feature at once, so the complementary test is as important as the recovery test: on a pure-noise map at the configured threshold, the extractor must find *nothing*. It does — zero supra-threshold time bins, zero duration, and `NaN` for every frequency extent — with the peak noise energy at $Z = 14.6$, comfortably below the threshold of 20. At the initially chosen $Z = 5.5$ the same map produced hundreds of supra-threshold bins.

4.4 A closure test on real triggers

Injections check the extractor against a signal built to the same model it assumes, which is necessary but circular. A weaker but non-circular check is available on real data: the features are over-determined, so they can be asked whether they agree with each other.

On the first 126 scattered-light events extracted from H1 O3a, the median measurements are an arch repetition period of 0.21 s, a harmonic comb spacing of 15.0 Hz, and a maximum arch frequency of 41.5 Hz. Those three are measured independently – from the autocorrelation of the envelope, from a frequency slice through the apex, and from the extent of the supra-threshold region. Taken together they imply

- a driving frequency $f_{\mathrm{drive}} = 1/2T_{\mathrm{arch}} = 2.4\,\mathrm{Hz}$, which falls in the anthropogenic band;
- a peak surface velocity $v = \lambda f_1^{\max}/2 = 8.0\,\mu\mathrm{m\,s^{-1}}$;
- a displacement amplitude $X_0 = \lambda f_1^{\max}/4\pi f_{\mathrm{drive}} = 530\,\mathrm{nm}$;
- $f^{\max}/f_1^{\max} = 2.8$ resolved harmonics.

Half a micron of optic motion at a few hertz, coupling through two or three bounce orders, is an unremarkable description of fast scattering — which is the point. Had the extractor been measuring an artefact of its own thresholds, there would be no reason for these three independent numbers to combine into a physically sensible displacement.

The arch period is measurable for 56 % of events and the harmonic spacing for 29 %; the remainder are too short or too faint for the autocorrelation and the comb finder respectively. That completeness is a property of the sample, not a failure, but it is why the classifier is built to treat a missing feature as informative rather than imputing it silently.

4.5 What only the real data revealed

Injections test the physics; they cannot test the archive. Three defects survived the entire simulated-signal suite and appeared within minutes of the first contact with live data. They are recorded because each one is a class of mistake that a validation-against-injections strategy is structurally blind to.

The cache never wrote anything. Both cache writers built a temporary file, wrote it, and renamed it into place. But `numpy.savez_compressed` appends `.npz` to any filename that does not already end in it, so the data landed in `<hash>.npz.tmp.npz` while the rename looked for

`<hash>.npz.tmp`. The very first real fetch died on it. No injection test touched the cache, so nothing caught it; the suite now round-trips both writers and checks that a failed write leaves nothing behind.

The catalogue counts triggers, and unevenly. Described in Section 3.4: the class-dependence is the part that matters, since 2.44 triggers per scattered-light event against 1.35 per control event is an 80 % relative over-representation of the very class under study. Nothing about a simulated arch train would have exposed this, because the simulator produces one event per event.

A fifth of the trial sample was in a hole. Twelve of the first forty events failed, all inside one 4096 s block that GWOSC does not serve — neither as a whole file nor as a narrow window within it. The error handling was already correct, in that the stage logged the failures and carried on; what was wrong was the cost, since every event in the dead block spent the full retry budget twice over. Unavailable blocks are now remembered, and the count of events lost to data gaps is reported with each stage rather than being silently absent from the sample.

Taken together these justify the shape of the suite: injections fix the physics and the estimator, and the first real run is treated as a test in its own right rather than as the start of production.

4.6 Test suite

The results above are encoded as assertions rather than left as a one-off check: 114 tests cover the feature contract, the recovery tolerances of Table 4, the null test, the calibration round trip, a negative control in which a predictor wrong by three decades must not still appear to agree, the GPS-time conversions against the published epoch boundaries, the segment arithmetic, the statistical estimators, and the two data-availability facts of Section 2. They require no network access and run in about three minutes.

5 Results

Status. The pipeline has been validated end to end against simulated injections (Section 4) and every data path has been verified against the live services, but the production run over the full O3 and O4a samples has not yet been executed. Quantities awaiting that run are shown as **[like this]**. They are placeholders, not measurements, and no value in this section should be quoted until this box disappears — it does so automatically once `paper/numbers.tex` is written by stage 06 of the pipeline. The structure below, the statistical treatment, and the caveats are final.

5.1 The O3 sample

Table 5 gives the size of the labelled sample after the cuts of Section 2.2.

5.2 Morphology of the O3 population

The distribution of the arch repetition period is the central descriptive result, because Eq. (4) converts it directly into a statement about the mechanical disturbance. Figure 3 shows it with the implied driving frequency on the upper axis.

Table 5: Gravity Spy triggers surviving the selection cuts, by detector and epoch, and the scattered-light fraction. Produced by `results/tables/catalog_summary.csv`.

Detector	Epoch	Triggers	Scattered light	Fraction	Median SNR
H1	O3a	[N]	[N]	[f]	[s]
H1	O3b	[N]	[N]	[f]	[s]
L1	O3a	[N]	[N]	[f]	[s]
L1	O3b	[N]	[N]	[f]	[s]



Figure 2: A representative slow-scattering trigger with the extracted arch envelope overlaid. The harmonic comb, the smooth rise and fall, and the regular repetition are the features the analysis quantifies.



Figure 3: Arch repetition period. The upper axis gives the implied driving frequency $f_{\mathrm{drive}} = 1/2T_{\mathrm{arch}}$; the microseismic band (0.1 to 0.3 Hz) and the anthropogenic band (1 to 6 Hz) can be read off directly. The dotted line at 1.5 s is the slow/fast split.

Table 6 decomposes each sample into the two populations. The columns headed "in band" are the check that the decomposition means what it claims: the slow population's implied driving frequencies should fall in the microseismic band and the fast population's in the anthropogenic band. If they do not, the split is a curve-fitting artefact rather than a physical separation.

Table 6: Slow/fast decomposition. "In band" is the fraction of each population whose implied driving frequency falls in the corresponding ground-motion band.

Detector	Sample	N	Slow fraction	Fast fraction	Slow in band	Fast in band
H1	O3	**[N]**	**[f]**	**[f]**	**[f]**	**[f]**
H1	O4a	**[N]**	**[f]**	**[f]**	**[f]**	**[f]**
L1	O3	**[N]**	**[f]**	**[f]**	**[f]**	**[f]**
L1	O4a	**[N]**	**[f]**	**[f]**	**[f]**	**[f]**

5.3 Auxiliary correlation in O3

5.3.1 Rate versus ground motion

Table 7 gives the Spearman correlation between the hourly scattered-light rate and the hourly band-limited RMS of each ground-motion channel, with six-hour moving-block bootstrap intervals.

Table 7: Rank correlation of the hourly scattered-light rate with ground-motion band-limited RMS, L1 O3b. Channel names are abbreviated: each is prefixed `L1:ISI-GND_STS_` and suffixed `.rms`. Intervals are 95 % moving-block bootstrap. Full table in `results/tables/aux_rate_correlations.csv`.

Channel	Band	ρ	95 % interval
`ITMY_Z_BLRMS_100M_300M`	microseism	**[r]**	**[ci]**
`ITMY_X_BLRMS_100M_300M`	microseism	**[r]**	**[ci]**
`ITMY_X_BLRMS_1_3`	anthropogenic	**[r]**	**[ci]**
`ITMY_X_BLRMS_3_10`	anthropogenic	**[r]**	**[ci]**
`ITMY_Z_BLRMS_30M_100M`	earthquake	**[r]**	**[ci]**

The strongest channel is **[channel]**, at $\rho =$ **[rho]**.

Three supporting checks accompany it. The top-to-bottom quartile rate ratio for that channel is **[ratio]**, which states the effect in operational terms: how much more often scattering occurs in the noisiest quarter of hours than in the quietest. The lag scan peaks at **[lag]** hours, against the zero expected for an instantaneous mechanical response. The partial correlation of rate with the anthropogenic band, controlling for the microseism, tests whether the two bands carry independent information or merely track one another.

5.3.2 The per-trigger predictor

Table 8 gives the fitted witness calibration and the agreement measured on the disjoint validation set. The calibration is quoted with the range over which the training objective stays within 1 % of its maximum: a narrow range indicates that the witness genuinely constrains the prediction, a wide one that it does not.



(a) Correlation summary.



(b) Rate against ground motion.

Figure 4: Left: Spearman ρ with moving-block bootstrap intervals for every ground-motion channel. Right: hourly rate against the band-limited RMS of the strongest channel, with binned medians and interquartile range.

Table 8: Scattering predictor $f_n = 2n|v|/\lambda$ against the witness `L1:SUS-OM3_M1_DAMP_L_IN1_DQ`. The calibration α is fitted on a training set and frozen before the validation measurement; the 1 % range is the interval over which the training objective stays within one per cent of its maximum.

Epoch	α (m/count)	1 % range	N_{val}	Median explained	Median r
O3a	[a]	[range]	[N]	[f]	[r]
O3b	[a]	[range]	[N]	[f]	[r]

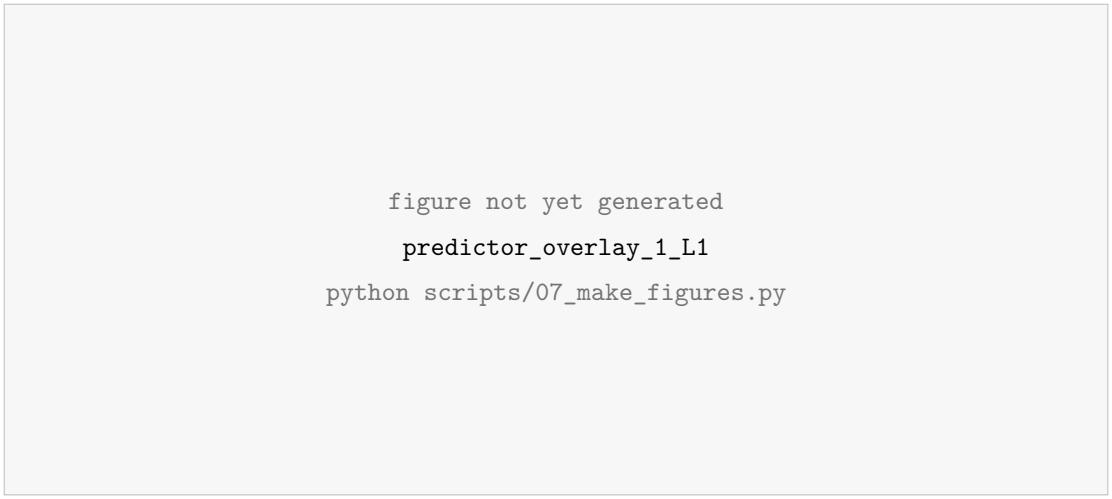


Figure 5: A real scattered-light trigger with the harmonic comb predicted from the auxiliary witness overlaid (dashed). The prediction uses only auxiliary data and the single fitted scale factor; the arches are from strain. Agreement between the two is the evidence that Eq. (3) describes the mechanism.

5.4 The O4a search

Table 9 records the coverage of the livetime-limited search and the number of candidates classified as scattered light. The identifier's cross-validated performance on held-out O3 folds is AUC = **[AUC]** with an efficiency of **[efficiency]** at a false-positive rate of 0.01 against the control classes — with the two caveats of Section 3.7 attached.

Table 9: Blind-search coverage and yield. Rates are per hour of *analysed* livetime; the coverage fraction is the analysed livetime as a fraction of the epoch's total.

Detector	Epoch	Windows	Analysed (h)	Coverage	Scattering	Rate (h^{-1})
H1	O3b	**[N]**	**[h]**	**[c]**	**[N]**	**[r]**
H1	O4a	**[N]**	**[h]**	**[c]**	**[N]**	**[r]**
L1	O3b	**[N]**	**[h]**	**[c]**	**[N]**	**[r]**
L1	O4a	**[N]**	**[h]**	**[c]**	**[N]**	**[r]**

5.5 O3 versus O4a

5.5.1 Rate

With the identical search applied to both runs, the ratio of the O4a rate to the O3 rate is **[ratio]** at H1 and **[ratio]** at L1, with Poisson intervals as given in Table 9.

5.5.2 Composition

The headline structural question is whether the slow, microseism-driven component was suppressed relative to the fast component. At H1 the slow fraction went from **[f]** in O3 to **[f]** in O4a, a change of **[change]**; at L1 it went from **[f]** to **[f]**, a change of **[change]** (95 % Newcombe intervals).

5.5.3 Morphology

Table 10 lists the features whose distributions differ most between the runs, ranked by Cliff's δ rather than by p-value, for the reason given in Section 3.9. The largest effect is in **[feature]**, at $\delta =$ **[delta]**.

Table 10: Feature-by-feature comparison, ranked by effect size. p-values are Benjamini–Hochberg corrected across all twenty features. Full table in `results/tables/feature_comparison.csv`.

Feature	Median O3	Median O4a	Cliff's δ	p_{adj}
[feature]	**[m]**	**[m]**	**[d]**	**[p]**
[feature]	**[m]**	**[m]**	**[d]**	**[p]**
[feature]	**[m]**	**[m]**	**[d]**	**[p]**
[feature]	**[m]**	**[m]**	**[d]**	**[p]**
[feature]	**[m]**	**[m]**	**[d]**	**[p]**

5.6 Systematics

Four checks bound the result.

Label purity. The Gravity Spy confidence cut is varied from 0.80 to 0.95. The O3 morphology medians move by **[shift]**, which bounds the effect of label contamination.



Figure 6: Morphology feature distributions for the O3 and O4a samples, normalised to unit area. Both samples are produced by the same search and the same classifier, so differences reflect the detector rather than the pipeline.

Trigger generator. The two comparison modes of Section 3.8 — like-for-like against the same search on O3, and against the Gravity Spy sample — are reported side by side. Agreement indicates the result does not depend on how triggers were found; disagreement is quantified and attributed.

Sampling. The livetime-limited search covers only **[coverage]** of O4a, so it is blind to structure on timescales longer than the sampling interval. The windows are drawn uniformly in livetime and stratified across the run, which makes the sample representative of the run as a whole but does not recover, for example, a two-week period of unusual behaviour.

Extraction bias. The 15 % high bias in maximum arch frequency (Section 4) applies identically to both runs and so cancels in the comparison. It is not corrected for, and no absolute statement is made using that feature.

6 Discussion

6.1 What the arch period buys

The single most useful property of scattered light for an analysis restricted to public data is that Eq. (4) makes the driving frequency a *strain-only* measurement. No auxiliary channel is required to say that a particular trigger was caused by motion at 0.15 Hz rather than at 2 Hz; the arch period says so directly.

That is what allows the O4a half of this study to exist at all. The O4a auxiliary release contains nothing that witnesses motion (Section 2.3.2), so an approach that depended on auxiliary data would simply stop at the end of O3. Reading the mechanism off the morphology does not.

The inference is only as good as the relation it rests on, which is why Section 5.3 matters beyond its own numbers. Establishing in O3 — where both the labels and the witnesses exist — that

the arch period does track ground motion, and that the predictor $f_n = 2n|v|/\lambda$ quantitatively explains individual arch trains, is what licenses reading an O4a arch period as a statement about the instrument's environment. Without the O3 auxiliary work, the O4a arch-period distribution would be a description of pictures rather than a measurement of physics.

6.2 Relation to previous work

Soni et al. [5] established the two-population picture in O3 using internal data, and identified relative motion between the main and reaction chains as the dominant slow-scattering path. Nandi et al. [11] report on scattered light at Livingston during O4, again finding two populations: one traced to microseismic ground motion and mitigated by installing baffles, the other to vertical ground motion coupling through a vacuum chamber and addressed by adding seismic isolation.

This work is not a substitute for either. It has access to a fraction of the channels, none of the engineering record, and no ability to intervene in the instrument. What it adds is a different kind of statement: a reproducible, externally checkable measurement of how the scattered-light population changed between the two runs, derived entirely from data anyone can download, with every intermediate step validated against signals of known parameters. Where its conclusions agree with the internal analyses, that agreement is independent evidence; where they disagree, the disagreement is informative about what public data can and cannot support.

The comparison with Nandi et al. [11] is the most direct test available. If the arch-period distribution measured here from O4a strain alone reproduces the two-population structure they identify from internal channels, the strain-only method is validated on real data as well as on injections.

6.3 Limitations

The O4a class definition is inherited from O3. Every O4a number rests on a classifier trained on O3 labels, so it presumes that the morphological boundary of "scattered light" did not move between runs. That presumption is not testable without O4a labels. The cross-validated efficiency and false-positive rate on held-out O3 folds (Section 5.4) are the quantitative statement of how well the transfer is expected to hold *within* O3; they are an upper bound on how well it holds across runs, not an estimate of it. A genuinely novel O4a glitch morphology resembling scattered light would be absorbed into the sample without comment.

The trigger generator is not Omicron. The excess-power finder of Section 3.8 is a stand-in. Its efficiency curve differs from Omicron's, in ways not characterised here. This is why the comparison is run like-for-like, with the same finder on both runs: the absolute rates are specific to this pipeline, and only the ratio between runs is meant to be compared with anything external.

Coverage. The O4a search covers a few per cent of the run. The sample is representative in the statistical sense — windows are drawn uniformly in livetime and stratified across the run — but it cannot resolve episodic behaviour, and a fortnight of anomalous scattering would appear only as a handful of windows.

One free parameter in the predictor. The counts-to-metres calibration is fitted, not derived, because GWOSC publishes auxiliary channels uncalibrated. The fit is honest — trained and validated on disjoint sets, with the constraint width reported — but a genuinely calibrated witness would turn the agreement statistic from a goodness-of-fit into a real test of Eq. (3).

Cross-validation is not grouped in time. Triggers minutes apart are not independent, so the quoted AUC is mildly optimistic. Time-grouped folds would be the correct treatment and are the obvious next refinement.

H1 has no public motion witness. The per-trigger predictor is a Livingston-only result. Whether the same calibration and the same explained fraction would be found at Hanford is untested, and untestable from public data.

6.4 What would strengthen this

In rough order of value per unit effort:

1. **An O4a trigger catalogue.** Even an unlabelled Omicron trigger list would remove the largest systematic here, by decoupling the trigger generator from this pipeline.
2. **Time-grouped cross-validation**, replacing the stratified folds with folds split on contiguous blocks of the run.
3. **Full O4a coverage**, which is a matter of compute rather than method; the pipeline is resumable and would scale to it unchanged.
4. **Direct comparison against Nandi et al. [11]** on overlapping times, which would test the strain-only classification against an independent, internally-derived identification.
5. **An O4a auxiliary release containing a motion witness**, which would make the O4a half of the study symmetric with the O3 half. The test suite is written to detect this if it happens.

7 Conclusions

This work characterises scattered-light transients in LIGO O3 and O4a using public data only, and asks how much of the physics survives that restriction.

1. **The morphology carries the mechanism.** Because the scattering frequency follows $|v|$ rather than v, the arch repetition period measures the driving motion directly, $f_{\mathrm{drive}} = 1/2T_{\mathrm{arch}}$, and the harmonic spacing at the apex measures the peak surface velocity. Both are strain-only measurements. Validated against injections, the arch period is recovered to better than 1 % and the harmonic spacing to within 7 %.
2. **Two public-data facts constrain what is possible, and neither is documented elsewhere.** No Gravity Spy classification catalogue has been released for O4a, and the O4a auxiliary release contains no seismometer, accelerometer or suspension witness. Together

they mean the O4a analysis must be strain-only, and that the auxiliary correlation is necessarily an O3 result. Both were established by direct verification before any code was written; the second is asserted in the test suite so that a future release would be detected.

3. **The O3 auxiliary release supports more than expected.** Alongside the complete ground-seismometer band-limited RMS family across the earthquake, microseismic and anthropogenic bands, it contains `L1:SUS-OM3_M1_DAMP_L_IN1_DQ` at 256 Hz — a genuine optic-motion witness, and the only full-rate one in either release. This makes a per-trigger test of $f_n = 2n|v|/\lambda$ possible from public data, which was not obviously the case at the outset.

4. **The O3 auxiliary work is what licenses the O4a interpretation.** Establishing in O3 that the trigger rate tracks ground motion, and that the predictor explains individual arch trains, is what allows an O4a arch period — measured with no auxiliary data at all — to be read as a statement about the instrument's environment rather than as a description of a picture.

5. **Validation against known signals changed the analysis, twice.** Capping the Q range corrected a factor-of-twenty error in the measured arch period of fast scattering, and raising the ridge threshold from $Z = 5.5$ to $Z = 20$ removed a noise floor that was saturating the frequency-extent features. Neither error was visible in the outputs; both were visible against injections. The residual 15 % high bias in maximum arch frequency is reported rather than tuned away, and is used only where it cancels.

The pipeline is public, resumable, and validated end to end. Its main extension — full O4a coverage — is a matter of compute rather than method, and the arrival of an O4a trigger catalogue would remove the largest remaining systematic without changing anything else.

Acknowledgements

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This work uses the Gravity Spy machine-learning classification data release [12] and the GWOSC O3 and O4 auxiliary channel releases [16, 17]. The Gravity Spy project is supported by the United States National Science Foundation and relies on the contributions of its citizen-science volunteers.

The analysis uses `gwpy` [19], `numpy` [26], `scipy` [27], `scikit-learn` [22] and `matplotlib` [28].

Data and code availability

All input data are public: strain and auxiliary channels from GWOSC, and the Gravity Spy classifications from Zenodo record 5649212. The analysis code, configuration, verbatim channel lists and decision log are at https://github.com/hardiksingh/LIGO_Glitch_Classification.

Every derived product is written with a provenance sidecar recording the configuration, the random seed and the software versions used to produce it.

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A The morphology feature vector

Each Q-scan is reduced to the twenty features below, computed by the function `extract_features` in `gwglitch.morphology`. All are derived from the normalised-energy map of Eq. (5), so none depends on the absolute calibration and all are directly comparable between detectors and between runs. A feature that is undefined for a given map is returned as `NaN` and never as a substituted value.

Notation: $Z(t, f)$ is the normalised energy, $Z_{\mathrm{thr}} = 20$ the ridge threshold, $f_{\mathrm{up}}(t)$ the highest supra-threshold frequency in time bin t (the arch envelope), and $E(f) = \sum_t Z(t, f)$ the supra-threshold energy marginal.

Feature	Units	Definition and purpose
Extent		
`snr_qscan`	—	$\sqrt{2Z_{\max}}$, the single-tile SNR of the loudest tile.
`energy_total`	—	Sum of supra-threshold Z. Spans orders of magnitude, so it is compared on a log scale.
`f_peak`	Hz	Frequency of the loudest tile.
`f_max`	Hz	99 % quantile of f_{up}. Biased high by $\sim 15\,\%$ because a Q tile of bandwidth f/Q extends above the true apex (Section 4); used only where that bias cancels.
`f_min`	Hz	1 % quantile of the lowest supra-threshold frequency.
`bandwidth`	Hz	`f_max` − `f_min`.
`duration`	s	Total time spanned by supra-threshold bins. Separates the seconds-long scattering trains from the millisecond blips.
Arch structure		
`n_arches`	—	Number of prominent maxima of f_{up}.
`arch_period`	s	T_{arch}, from the first prominent maximum of the autocorrelation of f_{up} (Section 3.3). The physically central feature: $f_{\mathrm{drive}} = 1/2T_{\mathrm{arch}}$.
`arch_period_scatter`	—	Normalised median absolute deviation of the peak-to-peak spacings. Small for a coherent train, large for irregular structure.
Harmonic structure		

Feature	Units	Definition and purpose
`n_harmonics`	—	Number of peaks in a frequency slice through the arch apex. Measures how many bounce orders are above threshold, and hence how non-linear the coupling is.
`harmonic_spacing`	Hz	Median spacing of those peaks, which equals the fundamental apex $2v_{\text{peak}}/\lambda$ and so measures the peak surface velocity directly.
Shape		
`ridge_slope_rms`	$\text{Hz}\,\text{s}^{-1}$	RMS of $\mathrm{d}f_{\text{up}}/\mathrm{d}t$.
`ridge_curvature_rms`	$\text{Hz}\,\text{s}^{-2}$	RMS of the second derivative. Arches are smoothly curved; whistles and lines are not.
`upward_downward_ratio`	—	Ratio of rising to falling time bins along the envelope. Close to unity for a symmetric arch train; asymmetric for classes with a fast rise and slow decay.
Energy distribution		
`time_symmetry`	—	Skewness of the energy time-marginal.
`frac_energy_below_40hz`	—	$\sum_{f<40} E / \sum E$. High for slow scattering.
`frac_energy_above_100hz`	—	$\sum_{f>100} E / \sum E$. Low for scattering, high for blips; one of the strongest discriminators against the control classes.
`spectral_centroid`	Hz	$\sum fE(f) / \sum E(f)$.
`band_concentration`	—	$1 - W_{1/2}/W_{\text{tot}}$, where $W_{1/2}$ is the width of the narrowest frequency band containing half the energy. Near 1 for narrow-band structure, near 0 for broadband.

Missingness is informative

Several features are genuinely undefined for some maps: `arch_period` when fewer than two arches are present, `harmonic_spacing` when fewer than two harmonic peaks are resolved, and `upward_downward_ratio` when the envelope never falls. That a feature could not be measured is itself diagnostic — a blip has no arch period because it has no arches — so the classifier imputes the median *and* carries a missingness indicator, rather than dropping the row or silently filling a number that would be indistinguishable from a measurement.

B Reproducibility

B.1 Environment

The analysis runs on Python 3.11 with `gwpy`, `numpy`, `scipy`, `pandas`, `scikit-learn`, `matplotlib` and `python-nds2-client`. The full specification is `environment.yml`:

```
conda env create -f environment.yml
conda activate gwglitch
pip install -e . --no-deps
```

Two aspects of that file are load-bearing and are recorded here because they cost real time to diagnose.

OpenBLAS is pinned. The unconstrained conda-forge solve selects MKL 2026.1.0, in which every LAPACK call terminates the interpreter with `STATUS_STACK_BUFFER_OVERRUN` (exit code `0xC0000409`) and no Python traceback whatsoever. `numpy.linalg.lstsq`, `numpy.linalg.solve`, `scipy.signal.detrend` and every `scikit-learn` regression die silently. `environment.yml` pins the OpenBLAS builds of `libblas`, `libcblas` and `liblapack`, and the environment check exercises the backend directly so that a recurrence is caught in a second rather than hours into a run.

`gwdetchar` is absent by design. Its dependency chain (`lalsuite`, `dqsegdb2`, `gwdatafind`) has no `win-64` build on conda-forge, so it cannot be installed in a native Windows environment. Everything the analysis needs from it — the Q-transform and the omega-scan rendering — is built on `gwpy`'s `q_transform`, which is pure Python and SciPy. The transform is the same algorithm [13]; only the plotting wrapper differs.

B.2 Verification before running

```
python scripts/00_check_environment.py
```

This checks package versions, exercises LAPACK and the SciPy filter design directly, parses the configuration and its channel lists, and confirms that GWOSC open strain, the NDS auxiliary server and the segment API all answer for both O3 and O4a. It exits non-zero on any mandatory failure. The run reported in Section 2 passed all six sections.

B.3 Pipeline stages

Each stage is a standalone program, resumable, and writes a JSON provenance sidecar next to its output recording the configuration path, the random seed, the software versions and the stage-specific parameters.

Stage	What it does
`00_check_environment.py`	Verifies packages, LAPACK, configuration and every remote service.

Stage	What it does
`01_fetch_gravityspy_catalog.py`	Downloads the four O3 CSVs (∼560 MB) from Zenodo 5649212, applies the selection cuts, and draws the time-stratified target and control sub-samples.
`02_run_qscans.py`	Q-scans each trigger and extracts the feature vector. The long stage: network-bound at roughly one trigger per second. Checkpoints every 100 triggers and skips completed work, so it can be interrupted freely.
`03_aux_correlation.py`	The O3 auxiliary analysis: hourly rate against ground-motion band-limited RMS with block-bootstrap intervals, quantile table, lag scan and partial correlation; plus the per-trigger predictor fitted on the L1 OM3 witness. Refuses to run on an O4 epoch.
`04_train_identifier.py`	Trains and cross-validates the strain-only classifier and fixes its operating point.
`05_search_o4a.py`	The livetime-limited blind search. Run once with `-epoch O4a` and once with `-epoch O3b` for the like-for-like comparison.
`06_population_comparison.py`	Rates, slow/fast decomposition, feature comparison; writes `paper/numbers.tex`.
`07_make_figures.py`	Renders every figure in this document to `results/figures`.

B.4 Determinism

Every random choice — the stratified catalogue sub-sampling, the search window placement, the cross-validation folds, the bootstrap resamples — draws from a seed set in `config/analysis.yaml` and recorded in each provenance sidecar. Re-running a stage with the same configuration reproduces its output exactly. Changing the seed changes the sample, and the size of that change is one of the systematics of Section 5.6.

B.5 Cost

Approximate wall-clock times on a laptop with a domestic connection:

Stage	Bottleneck	Approximate time
01 catalogue download, 560 MB	network	5 to 15 min
02 Q-scans, 400 blocks, ∼5600 events	network	15 to 40 h
03 auxiliary correlation	network	2 to 6 h
04 classifier training	CPU	2 to 5 min
05 O4a search, 400 windows	network and CPU	4 to 10 h
06 comparison	CPU	1 to 3 min
07 figures	mixed	5 to 20 min

These are measured, not estimated. The dominant cost is stage 02, and it is set by the number of GWOSC *file* downloads rather than by the number of events: open strain is served in 4096 s HDF5 files, and a request for any sub-interval fetches the whole file, timed at two to four minutes each on a domestic connection. The block cache described in Section 3.4 turns that from one download per event into one per block; the measured yield over a real run is about ten events per download, against a naive 1.4.

Every remote fetch is cached on disk keyed by a hash of the request, so a re-run after an interruption costs only the work not yet done – which matters, because a job of this length will be interrupted. The cache is the dominant disk consumer, at roughly 1 MB per 64 s strain window.

B.6 Tests

```
pytest
```

114 tests, no network access required, about three minutes. They cover the feature contract, the recovery tolerances of Table 4, the pure-noise null test, the witness-calibration round trip, a negative control on the predictor, the GPS conversions against the published epoch boundaries, the segment arithmetic, the statistical estimators, and the two data-availability facts of Section 2 — so that a future GWOSC release adding an O4a motion witness would announce itself as a test failure.

B.7 Building this document

The `paper` directory compiles standalone on Overleaf with pdfLaTeX and BibTeX. `\graphicspath` declares both `../results/figures` and `./figures`, so the source works whether it sits inside the analysis repository or has been uploaded on its own; in the latter case, copy `results/figures/*.pdf` into `paper/figures`.

Numbers awaiting the production run appear as `\result{}` placeholders, rendered in red so that no unfilled value can be mistaken for a measurement. Stage 06 closes the loop in two ways. It writes `paper/numbers.tex`, which redefines the inline macros to the measured values and removes the status notice at the head of Section 5; and it writes `paper/generated/*.tex`, the body rows of each results table, which the section files pull in through `\generated` when they exist and fall back to placeholders when they do not. The column specifications, captions and prose stay under human control; only the numbers are machine-written, and they cannot drift from the analysis that produced them.

The macro names are generated mechanically from the pipeline's keys, so `slow_fraction_o4a_h1` becomes `\ResultSlowFractionOFouraHOne` — digits are spelled out because TEX forbids them in a control sequence. `tests/test_latex.py` asserts that every key the pipeline emits has a declaration in `main.tex` and that every macro used in a section is declared, so the two halves cannot silently fall out of step.